

MACHINE LEARNING II

CUSTOMER SEGMENTATION

GROUP 9

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1 EXECUTIVE SUMMARY

1.1 BRIEF OVERVIEW OF THE PROBLEM

In today's competitive business environment, **understanding customer behaviour** is essential to gain a **strategic advantage**. Traditional marketing approaches often rely on generalized assumptions and predefined customer categories, which can overlook important behavioural patterns.

This project addresses the challenge of identifying natural groupings within a customer base **using unsupervised machine learning techniques**. By segmenting customers based on their characteristics and purchasing behaviour, businesses can create tailored marketing strategies that increase customer satisfaction, loyalty, and overall business performance. **Our goal** was to **discover these segments and propose targeted promotions** that **best align** with **each group's** preferences and needs.

1.2 SUMMARY OF METHODOLOGY AND APPROACH

To tackle the customer segmentation problem, we followed a structured and iterative data science workflow. The entire **development process was managed using GitHub**, where we collaborated efficiently and maintained a well-organized repository (further detailed in [Section 1.3](#)).

The initial phase of the project entailed an **examination and comprehension of the two datasets** provided, designated *customer_info* and *customer_basket*. These datasets encompass demographic, behavioural, and transactional data. Following this, a comprehensive data **preprocessing framework** was implemented in *customer_info*, addressing issues such as missing values, outliers, encoding, and data standardisation to **ensure data quality and consistency**.

Once the data was cleaned, we experimented with **multiple clustering algorithms** to identify the **most meaningful customer segments**. After testing different models, one got **selected** as it provided the **most interpretable clusters**. Our **choice was driven not only by technical performance but also by the clarity it offered in distinguishing customer profiles**.

Post-clustering, **each segment's unique features** were analysed to **uncover meaningful behavioural trends**. This analysis allowed us to better understand each customer group and served as the **foundation for designing targeted marketing strategies**.

Finally, using insights from both, the **clustering results** from *customer_info* and **association rules** derived from the *customer_basket* data, we proposed **personalized promotional campaigns tailored** to each segment. These campaigns **aim to boost engagement and align business goals with customer preferences**.

1.3 REPO STRUCTURE

As previously mentioned, *GitHub* was used to develop this project. To ensure modular development and maintain a clean and organized workflow, we structured our repository as follows.

1.3.1 FILES' ORGANIZATION

1. **Notebook Testing**
Contains Jupyter notebooks used during each development phase – preprocessing, clustering and analysis.
2. **Source Code**
Includes validated Python scripts containing the core functions used throughout the project.
3. **Clustering Outputs**
Stores outputs generated from the clustering processes.
4. **Datasets** (at root level)
Contains the original and pre-processed datasets used in the project.

1.3.2 BRANCHES' ORGANIZATION

To facilitate **parallel development and task separation**, we implemented a topic-based branching strategy (preprocessing and clustering). Within each topic branch, team members worked on their individual sub-branches. Once the contributions were tested and finalized, they were merged into the **main branch**, which served as the **project's single source of truth**.

2 EXPLORATORY DATA ANALYSIS AND PRE-PROCESSING

2.1 DATA SOURCES

Initially, the project is grounded in two key datasets: *customer_info* and *customer_basket*.

Before proceeding with preprocessing, an exploratory investigation was conducted on both datasets. This included checking the structure and shape of the data, identifying data types, reviewing summary statistics using *.describe()*, and assessing the presence of missing values. This initial phase helped us understand the contents and quality of the data, guiding our preprocessing and modelling strategies.

- ***customer_info*** contains detailed demographic and behavioural data for each customer. It includes variables such as number of children and teenagers at home, geographic location, spending habits across various product categories (e.g., groceries, electronics, meat, fish), loyalty card usage, and historical purchase behaviour. This dataset has **34,060 rows and 25 columns, with each row representing a unique customer**.
- ***customer_basket*** provides transactional data, containing **100 000 randomly sampled baskets**. Each entry includes the *customer_id*, *invoice_id*, and a list of goods purchased during a **specific transaction**. This dataset allowed us to analyse purchasing combinations and later mine association rules for targeted promotions.

2.2 DATA PREPROCESSING

2.2.1 GENERAL CORRECTIONS

In the initial review of the data, we identified several **areas for improvement** in the *customer_info* dataset. Key corrections included:

- **Splitting *customer_name* into two separate columns**: one for the actual name and another for the education level prefix (e.g., "PhD.", "BSc."), as values like "PhD. Curtis Tharp" were present in the original data.
- Converting ***customer_birthdate* to datetime format** and **deriving** additional features such as **age** (on the 9th of June), **birth month**, **day**, and **year** to retain as much temporal information as possible at this stage.
- Enriching customer behaviour by **calculating the number of distinct invoices and unique products purchased**, using the *customer_basket* dataset (a sample used primarily for verifying inconsistencies).
- **Creating new features**, such as a **binary variable indicating loyalty card ownership** (replacing the *loyalty_card_number*, which was not analytically useful), and **estimating *customer_lifetime*** based on the year of their first transaction.
- **Dropping redundant or unnecessary columns** – such as *customer_name*, *customer_birthdate*, *loyalty_card_number*, and *year_first_transaction* – after extracting all relevant information.

These corrections made sure that the data was in the right format for preprocessing.

2.2.2 DUPLICATES

To ensure data integrity, we **verified the presence of duplicate** entries using the **customer_id** as a **unique identifier**. After running the check, we confirmed that no duplicate records existed in the dataset. As a result, **no further action was needed** for this step.

2.2.3 INCONSISTENCIES

During our data validation process, **several inconsistencies** were **identified and corrected** in the *customer_info* dataset. These inconsistencies ranged from **negative values** where they are not logically possible, to **incorrect proportions** or **mismatches between related variables**.

The table below summarizes the main issues detected and the corresponding corrections applied:

Inconsistency	Description	Treatment Applied
Negative <i>kids_home</i> , <i>teens_home</i> , <i>distinct_stores_visited</i>	These variables should only contain non-negative integers	Converted to positive values
Negative <i>number_complaints</i>	A customer cannot have negative complaints	Replaced with 0
Negative <i>customer_lifetime</i>	Lifetime cannot be negative	Replaced with 0
<i>distinct_products_sum</i> higher than <i>lifetime_total_distinct_products</i>	A count from a sample can't be higher than the total.	Set total to match the higher value
<i>lifetime_total_distinct_products</i> ≤ 0	Indicates no product purchases or invalid value	Replaced with 1 or made positive
<i>percentage_of_products_bought_promotion</i> < 0 or > 1	Should be a proportion between 0 and 1	Clipped to range [0, 1]
Negative <i>lifetime_spend_*</i> columns	Spending cannot be negative	Converted to positive values
Age at first transaction < 18	Unlikely a customer started shopping before 18	Set <i>customer_lifetime</i> to 0

2.2.4 UNIVARIATE OUTLIER DETECTION

An exploratory analysis was performed using **interactive visualisations** with **Plotly**, with a view to gain a **better understanding of the distribution** of variables and **identifying potential outliers**.

For **numerical variables**, we used box plots and histograms to detect univariate outliers based on their spread and distribution.

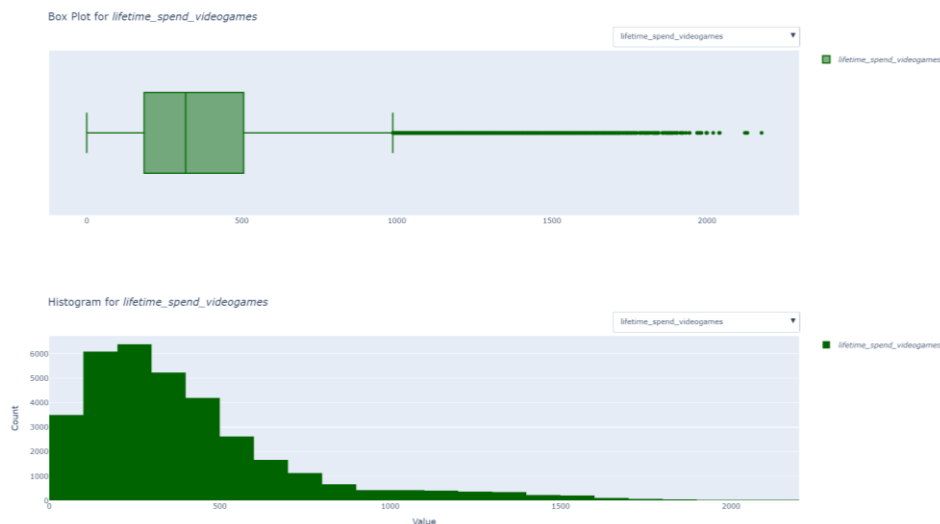


Figure 1 - Variables' Distribution

For features that exhibit **categorical behaviour** – whether **truly categorical or numerical variables with discrete limited values** (e.g., *kids_home*, *number_complaints*, *typical_hour*) – we used bar charts instead. This approach **helped avoid misleading interpretations** and provided a **clearer view of the distribution of discrete values**. These visualizations enabled us to assess value frequencies and detect possible anomalies within individual features.

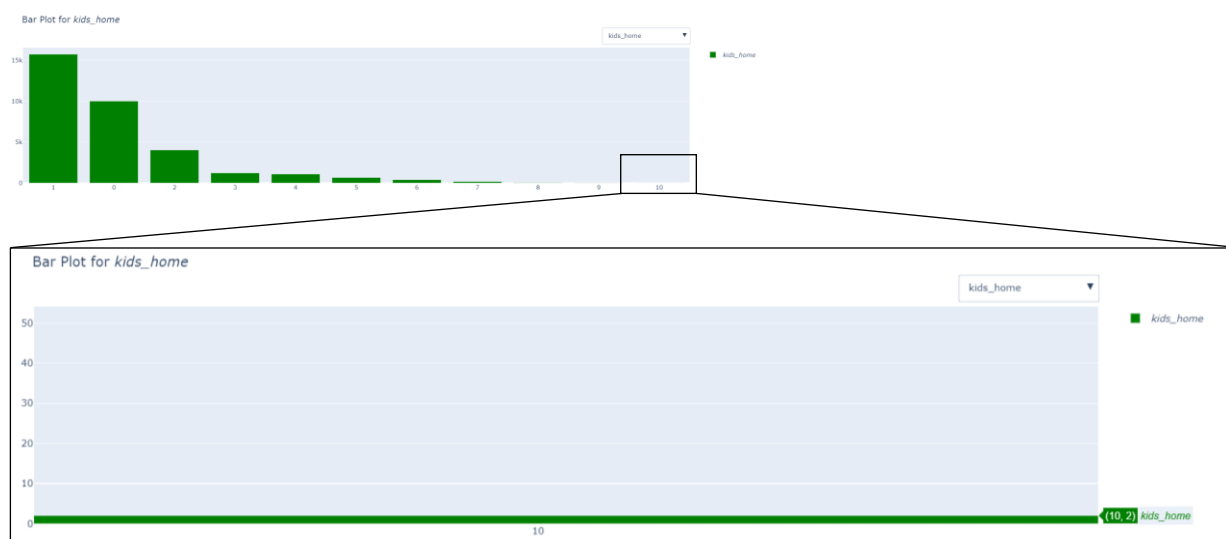


Figure 2 - Outlier Detection

Following a visual identification of potential outliers, a **rule-based filtering approach** was applied. For selected features, **upper thresholds were defined** based on empirical distribution analysis. Data points that **exceeded these thresholds** were identified as **outliers**. Instead of deleting them, we decided to **move these records to a separate dataset** for potential future reference or analysis.

This separated dataset will also undergo the subsequent preprocessing steps to ensure consistency and enable comparative analysis if needed later on.

2.2.5 MISSING VALUES

Before applying automated imputation techniques, a series of domain-informed rules to handle missing values in key features were implemented. These rules were guided by **common sense and business logic** to **ensure meaningful imputations**:

- **kids_home** and **teens_home**: logical assumptions were applied to impute missing values **based on the relationship between the two** features.
 - If **kids_home** was missing but **teens_home** had a non-zero value, we inferred that the household had no younger children and **set kids_home to 0**. The **opposite was also applied**.
 - When **both values were missing**, we assumed the **household had no dependents** and imputed both as 0.
- **education_level**: missing values in this column were imputed based on the customer's age, using a rule consistent with the **minimum required education levels** defined by the **Portuguese education system**. The age-based mapping was as follows:
 - $\text{age} \geq 59 \rightarrow \text{"4th"}$
 - $\text{age } 44 \text{ to } 58 \rightarrow \text{"6th"}$
 - $\text{age } 30 \text{ to } 43 \rightarrow \text{"9th"}$
 - $\text{age} \leq 29 \rightarrow \text{"Hs" (High School)}$

This approach provided a contextually sound estimation of educational attainment, **ensuring coherence with demographic expectations**. The mapping logic was inspired by the official guidelines available at <https://edu.azores.gov.pt/seccoes/matriculas-escolaridade-obrigatoria/>.

2.2.6 ENCODING

In this step, the **only categorical features requiring encoding** were **education_level** and **customer_gender**. These were transformed using **predefined mappings**. For the **gender** variable, **1** represents **female** and **0** represents **male**. Regarding **education level**, the categorical values were **replaced with the corresponding number of years** of schooling

This encoding was crucial to **ensure compatibility** with the subsequent **scaling** process and particularly **important for clustering algorithms**, which operate more effectively with fully numerical inputs.

2.2.7 SCALING

After encoding, **all features were scaled** to ensure consistency across the dataset. This step is particularly **important for distance-based techniques** – such as clustering and KNN imputation – where differences in feature scales can significantly distort the results.

It was applied **MinMax Scaler**, in both datasets, which transforms feature values into a [0, 1] range. This method was chosen for its simplicity and ease of interpretation post-scaling.

The **customer_id** column was **excluded from this transformation**, as it functions **solely as an identifier** and does not contribute to analytical computations.

2.2.8 MISSING VALUES (WITH KNN)

To definitively address the remaining missing values, we implemented **K-Nearest Neighbours** (KNN) imputation. This technique estimates missing values based on the values of the k most similar instances – in our case, the 5 nearest neighbours as it **offers a good balance between bias and variance**. Using fewer neighbours may introduce noise, while too many could smooth out important local patterns. This choice ensures a more context-aware and data-driven imputation process.

KNN imputation preserves the underlying relationships in the dataset by filling missing values according to observed patterns rather than arbitrary constants. Subsequent to applying this method, all missing values were successfully filled, resulting in a complete and consistent dataset, ready for the following steps in preprocessing and analysis.

2.2.9 MULTIVARIATE OUTLIER DETECTION

As aforementioned in [Section 2.2.4](#), we had **already identified univariate outliers**. However, it is **equally important to detect the multivariate ones**, which may not be evident when considering features individually. To address this, we employed the **DBSCAN** (Density-Based Spatial Clustering of Applications with Noise) algorithm.

DBSCAN is **well-suited** for outlier detection in multidimensional data because **it identifies as outliers the data points that do not belong to any dense region** (i.e., not assigned to any cluster). **These isolated points** are likely to **represent anomalies or noise** in the data.

One **challenge** of using DBSCAN is selecting **appropriate** values for its **two key parameters**:

- **eps**: the maximum distance between two data points for them to be considered as part of the same neighbourhood.
- **min_samples**: the minimum number of data points required to form a dense region.

For **min_samples**, we followed a common heuristic: setting it **equal to the number of features** (excluding the **customer_id** column), which in our case is **29**.

To choose a suitable **eps** value, we **tested a range from 0.5 to 1.2** (inclusive), using a **step of 0.1**. For each value, we evaluated the **number of clusters** formed and the **total number of observations labelled as outliers**. The results are summarized below:

The **best balance** between preserving data and effectively identifying noise was found at $\text{eps} = 1.0$. At this setting, **67 outliers** were detected by DBSCAN, therefore we **removed** those from the *customer_info* dataset and **stored them in the separate excluded rows dataset** for potential future analysis.

```
=====
With eps = 0.5
  Number of clusters (not outliers): 6
  Total number of outliers: 30652
=====
With eps = 0.6
  Number of clusters (not outliers): 17
  Total number of outliers: 28961
=====
With eps = 0.7
  Number of clusters (not outliers): 18
  Total number of outliers: 20707
=====
With eps = 0.7999999999999999
  Number of clusters (not outliers): 14
  Total number of outliers: 8558
=====
With eps = 0.8999999999999999
  Number of clusters (not outliers): 4
  Total number of outliers: 1065
=====
With eps = 0.9999999999999999
  Number of clusters (not outliers): 4
  Total number of outliers: 67
=====
With eps = 1.0999999999999999
  Number of clusters (not outliers): 1
  Total number of outliers: 6
=====
With eps = 1.1999999999999997
  Number of clusters (not outliers): 1
  Total number of outliers: 0
=====
```

Figure 3 - Hyperparameters for DBSCAN

2.2.10 REDUNDANCY

As part of a feature selection strategy, we also considered the **redundancy and correlation between variables**. This is important to ensure that the **clustering algorithm is not influenced by duplicated, highly correlated or deemed unnecessary information**, which can **distort the clustering structure and reduce model interpretability**.

To detect redundancy, we computed a **correlation matrix** to assess the relationships between numerical features. This analysis revealed several strong correlations, indicating that some variables carried overlapping information.

Based on this, we removed the following columns:

- *birth_month*
- *birth_year*
- *distinct_invoice_sum*
- *distinct_products_sum*

By eliminating them, we improved the dataset's quality and reduced noise, ensuring that the remaining variables offer more distinct and meaningful signals for the clustering algorithms applied in the succeeding steps.

2.2.11 EXPORT DATA

After completing the **full preprocessing**, we **exported** both **scaled** and **non-scaled** versions of the **customer_info** and **excluded_ids** datasets. This dual export strategy ensures the data is preserved in formats suitable for various analytical purposes, particularly the upcoming clustering stage focused on customer segmentation.

The **scaled** dataset includes features that have been normalized using **MinMax** scaling. This transformation is **crucial for clustering algorithms like K-Means, which rely on distance metrics** and are highly sensitive to the scale of input features. Scaling ensures that all variables contribute equally to the clustering process. In contrast, the **non-scaled dataset** retains the original feature distributions post-cleaning and transformation. This version is particularly **valuable for interpretability, reporting, or visual inspection, where maintaining the original data context may be more informative.**

3 CUSTOMER SEGMENTATION AND CLUSTERING

3.1 SEGMENTATION APPROACH

3.1.1 CLUSTER MAKRO

This second phase starts by **examining the geolocations** using the provided latitude and longitude columns, we observed **a cluster** of values concentrated around the **Makro area**.

After **performing outlier detection** and removal, these **Makro-related entries were no longer present** in the dataset, indicating that they had been classified as outliers, which seems plausible by looking at the map.

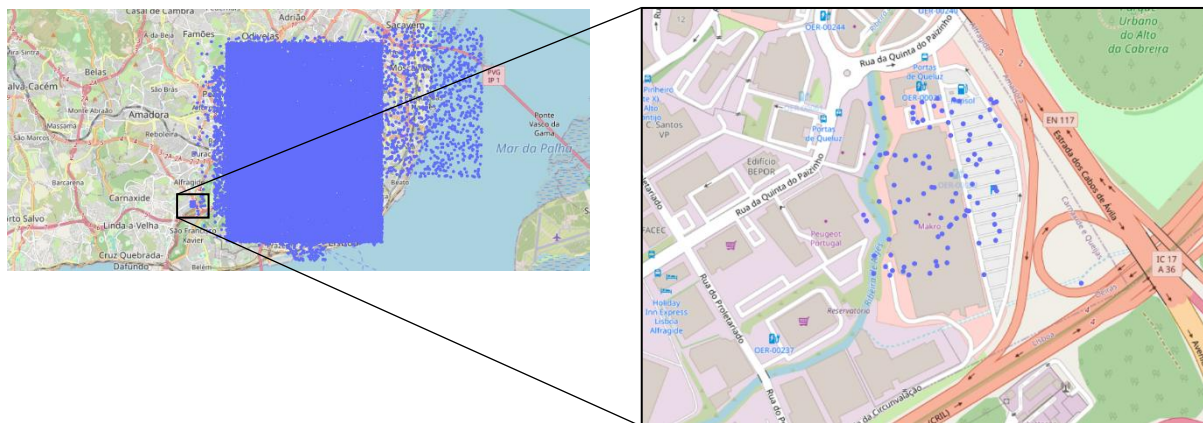


Figure 4 - Makro Cluster

To prevent them from **skewing the clustering analysis**, we isolated this cluster.

3.1.2 KEPT COLUMNS

Before applying clustering algorithms, we identified **certain features that could negatively impact the clustering process** and decided to **ignore them during model building**, while still **keeping them in the dataset for later interpretation**. These features include:

- **customer_id**: acts solely as a unique identifier and carries no analytical meaning.
- **gender** and **has_loyalty_card**: binary variables that can **disproportionately influence** distance-based clustering methods.
- **latitude and longitude**: a cluster based on geographical coordinates was previously identified. **Including these variables again would risk reinforcing that spatial separation unnecessarily.**

This approach preserves valuable contextual information that may enhance the interpretability of the final segmentation.

3.1.3 ASSIGNING OUTLIERS TO A CLUSTER

With clustering applied to the cleaned dataset (i.e., with outliers removed), we calculated the **distance from each previously excluded outlier to all cluster centroids obtained** from the clustering solution. Each outlier was then assigned to the nearest cluster based on these distances, ensuring all customers were ultimately included in the segmentation for interpretation purposes.

We used the **non-scaled datasets** for this step because the dataset used for clustering had been scaled independently from the dataset containing the outliers (Section 2.2.7). **Using the scaled version in this case could have introduced bias into the distance calculations and thus affected the cluster assignment results.**

3.1.4 CLUSTERING EVALUATION AND INTERPRETATION

To evaluate and interpret each clustering solution, the following tools were operated, before and after assigning the outliers to a cluster:

- **Cluster Mean Matrix (Scaled):** Helps identify dominant features in each cluster.
- **Cluster Mean Matrix (Original Values):** Allows for practical interpretation of characteristics.
- **UMAP Visualization:** Visually assesses the separation and cohesion of clusters.
- **Cluster Size Distribution:** Ensures clusters are reasonably balanced.
- **Silhouette Plot:** Assesses the quality of clustering by measuring how similar an observation is to its own cluster compared to others.

All visualizations presented in the following sections correspond to the final clustering solution (already with outliers assigned, Section 3.1.3) to each respective method.

3.1.5 HIERARCHICAL CLUSTERING

Initially, we implemented **agglomerative hierarchical clustering without specifying the number of clusters**. A **dendrogram** was plotted to visualize how clusters were formed and merged at each step.

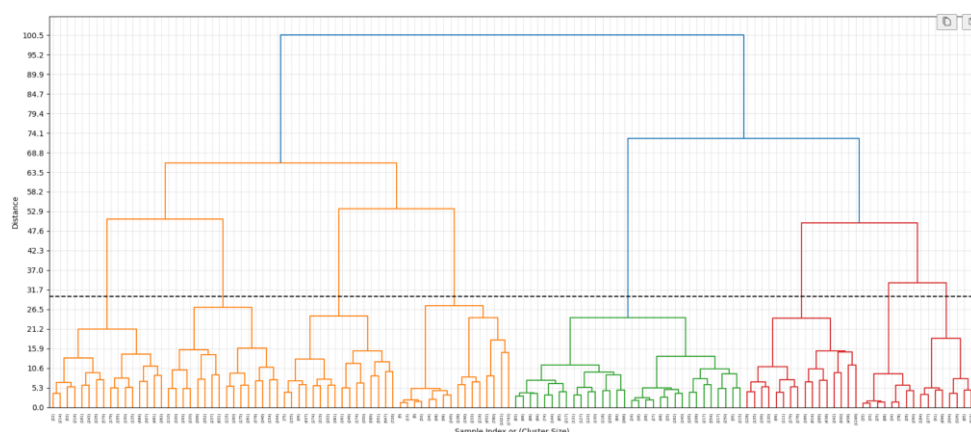


Figure 5 - Hierarchical's Dendrogram

As soon as the dendrogram's structure was carefully analysed, we determined that the **optimal number of clusters is 9**. Then the algorithm was re-applied considering this value, the outliers got associated and the final clustering solution was obtained.

Final Outputs

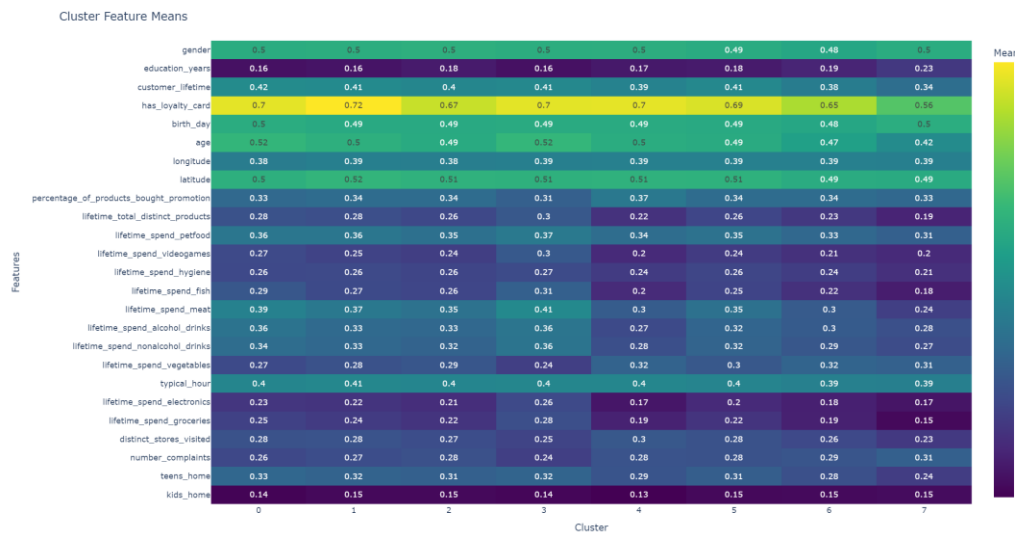


Figure 6 - Hierarchical's Scaled Mean Matrix

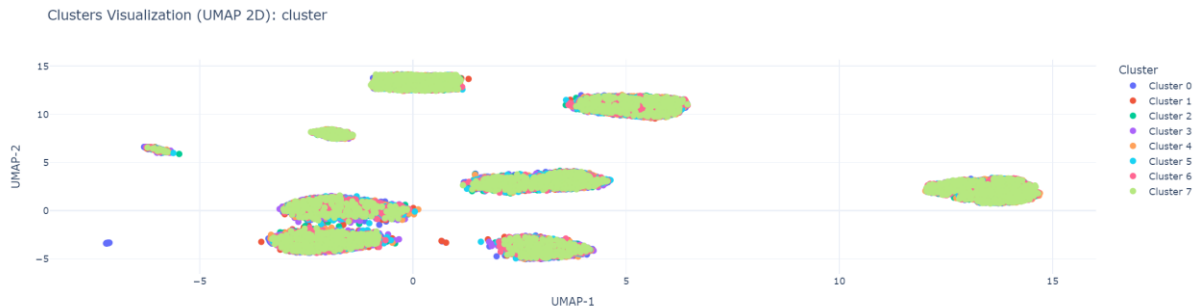


Figure 7 - Hierarchical's UMAP

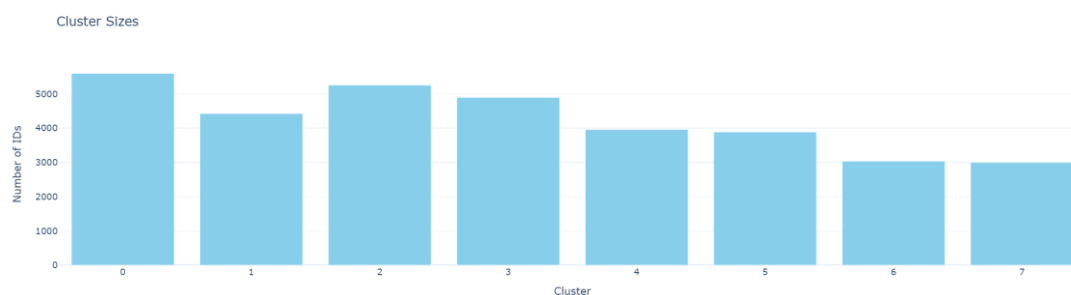


Figure 8 - Hierarchical's Cluster Sizes

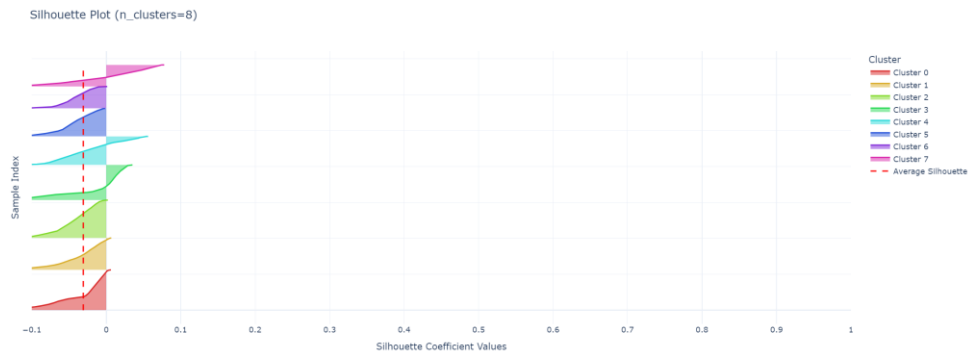


Figure 9 - Hierarchical's Silhouette

The figures show that the algorithm yielded **poor results**.

3.1.6 K-MEANS CLUSTERING

To identify the **optimal number of clusters for K-Means**, we first plotted the **Elbow Method**, which measures the within-cluster sum of squares (WCSS) against the number of clusters.

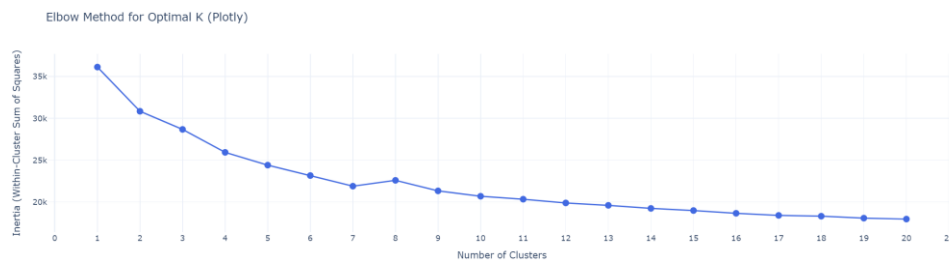


Figure 10 - K-Means' Elbow Chart

The plot suggested that the ideal number of clusters was **10**. Using this value, we implemented the K-Means algorithm, distributed the outliers and generated the respective cluster assignments.

Final Outputs

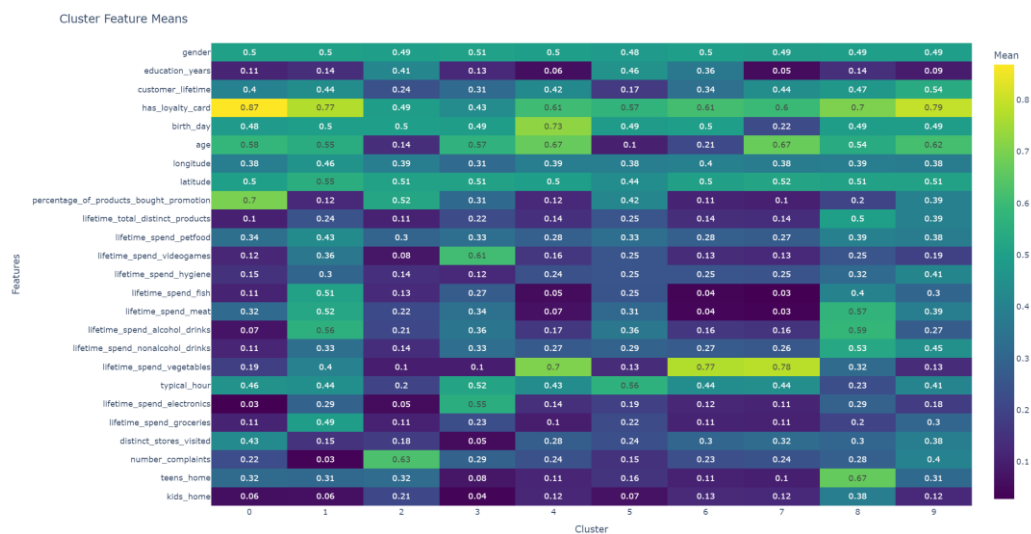


Figure 11 - K-Means' Scaled Mean Matrix

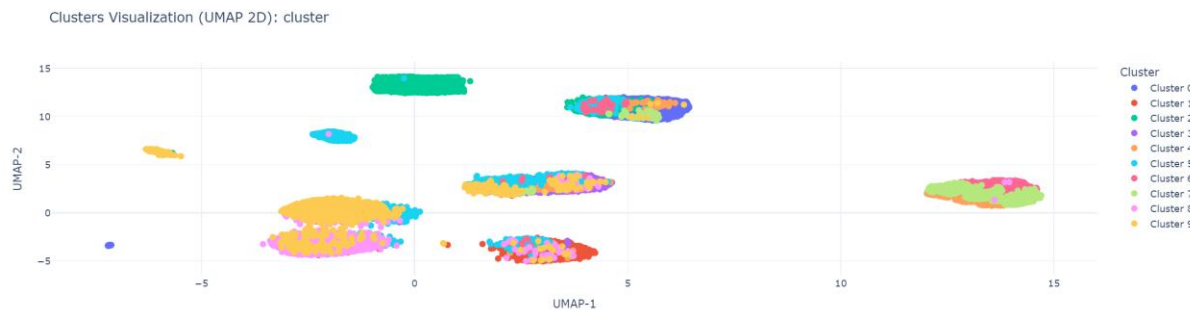


Figure 12 - K-Means' UMAP

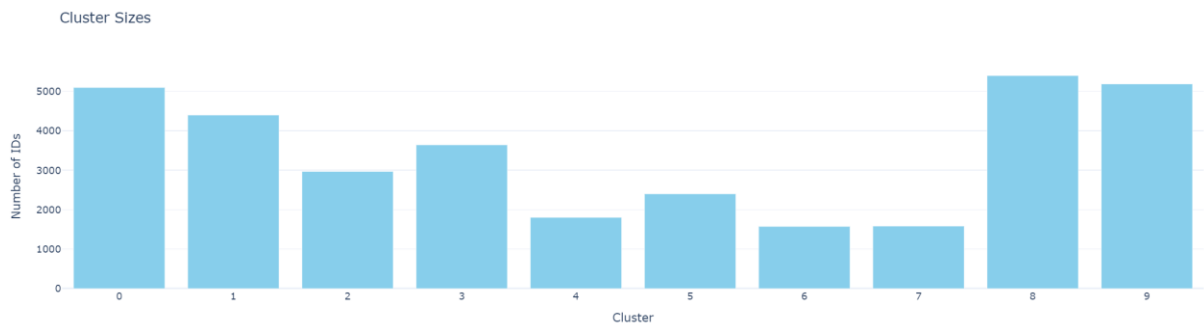


Figure 13 - K-Means' Cluster Sizes

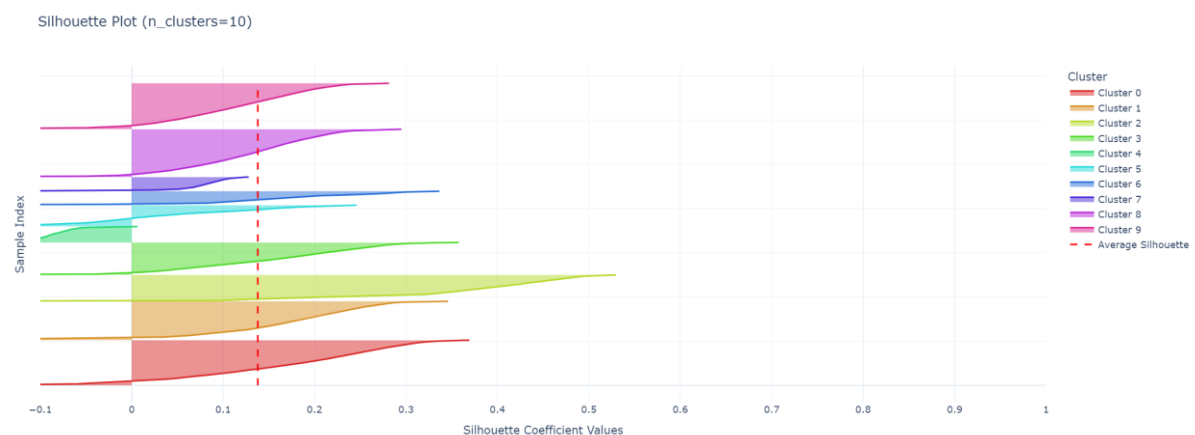


Figure 14 - K-Means' Silhouette

According to the figures above, this algorithm demonstrates **sufficient performance to be considered a candidate for the champion model**.

3.1.7 K-MEANS + HIERARCHICAL CLUSTERING

This hybrid approach was selected to **combine the interpretability of hierarchical clustering with the efficiency of K-Means**.

We began by using the **Elbow Method** to define a relatively **high number of K-Means clusters** (e.g., 20), ensuring a granular separation of the data.

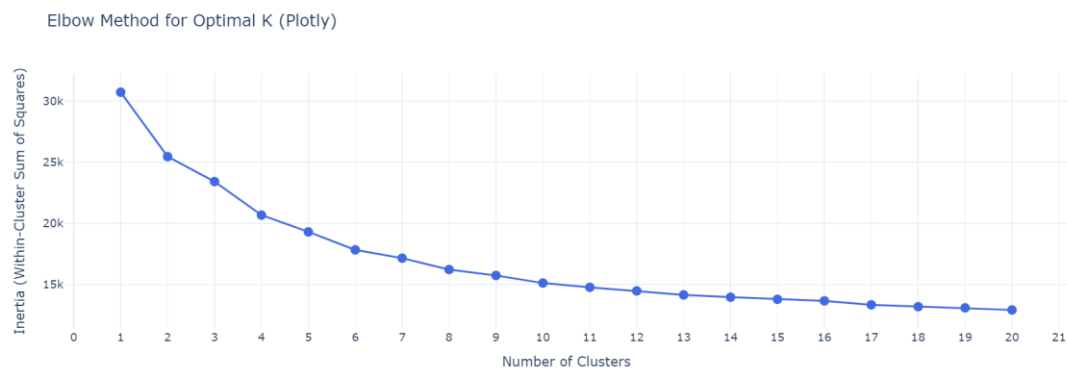


Figure 15 - K-Means' Elbow Chart

Next, we computed the **centroids of these K-Means clusters** and **employed hierarchical clustering** to them, allowing us to analyse which centroids were more similar. A dendrogram was plotted based on the **distances between these centroids**.

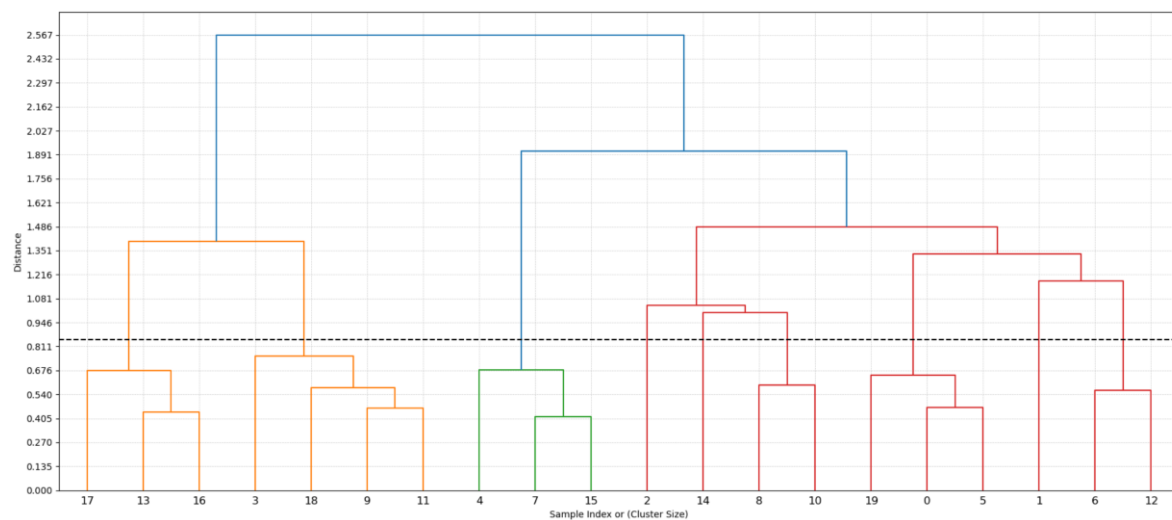


Figure 16 - Hierarchical's Dendrogram

It suggested 9 main clusters, so we **grouped the original K-Means clusters** accordingly by **reassigning each K-Means cluster to its respective hierarchical cluster**:

kmeans	hierarchical	
0	9	0
1	3	0
2	18	0
3	11	0
4	17	1
5	16	1
6	13	1
7	4	2
8	15	2
9	7	2

10	6	3
11	12	3
12	8	4
13	10	4
14	0	5
15	5	5
16	19	5
17	1	6
18	2	7
19	14	8

Figure 17 - K-Means' Clusters to Hierarchical's Clusters

As we have the clusters well-defined, the outliers' observations got assigned, which resulted in a final, merged clustering solution.

Final Outputs

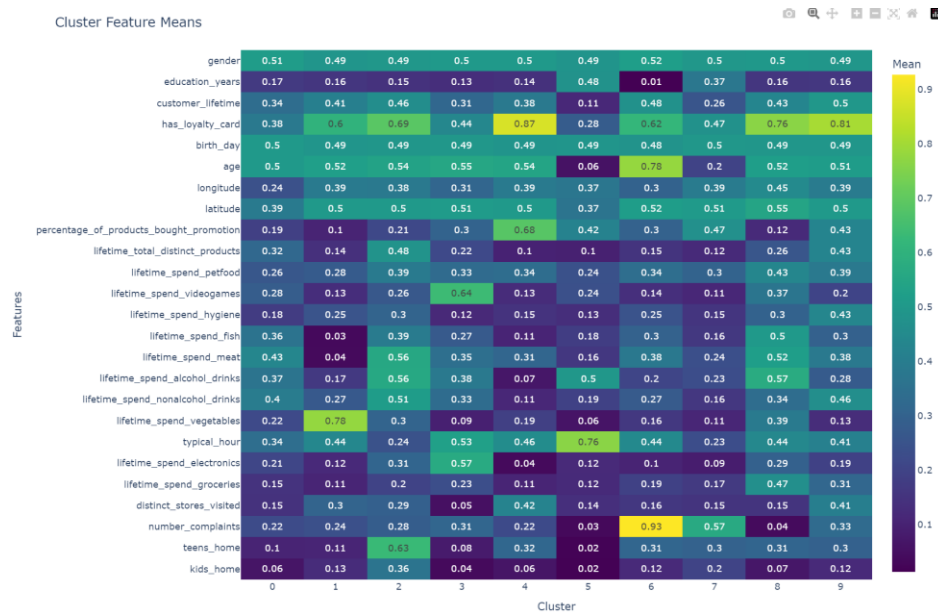


Figure 18 - K-Means + Hierarchical Scaled Mean Matrix

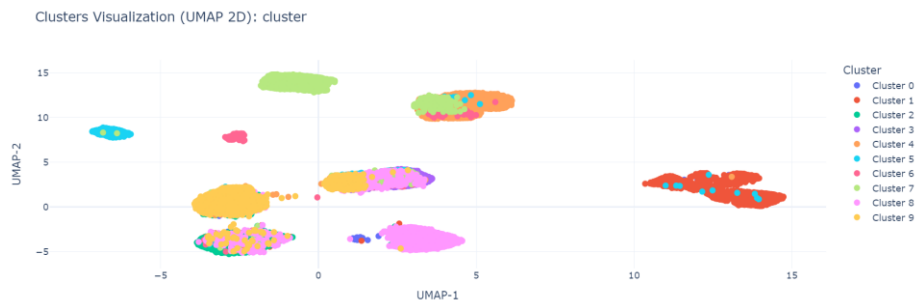


Figure 19 - K-Means + Hierarchical UMAP

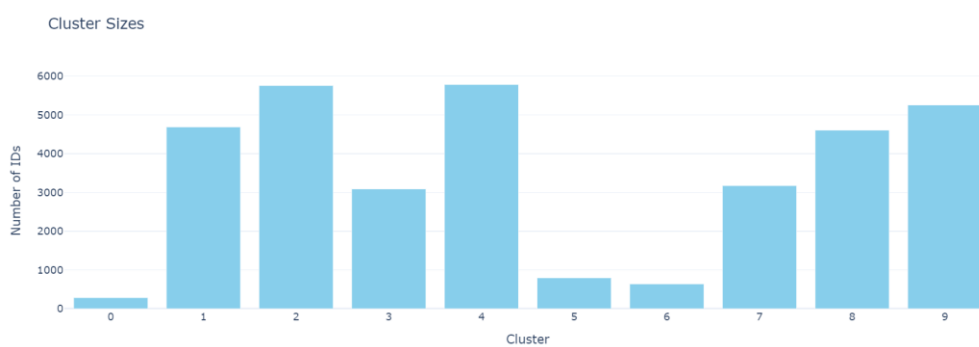


Figure 20 - K-Means + Hierarchical Cluster Sizes

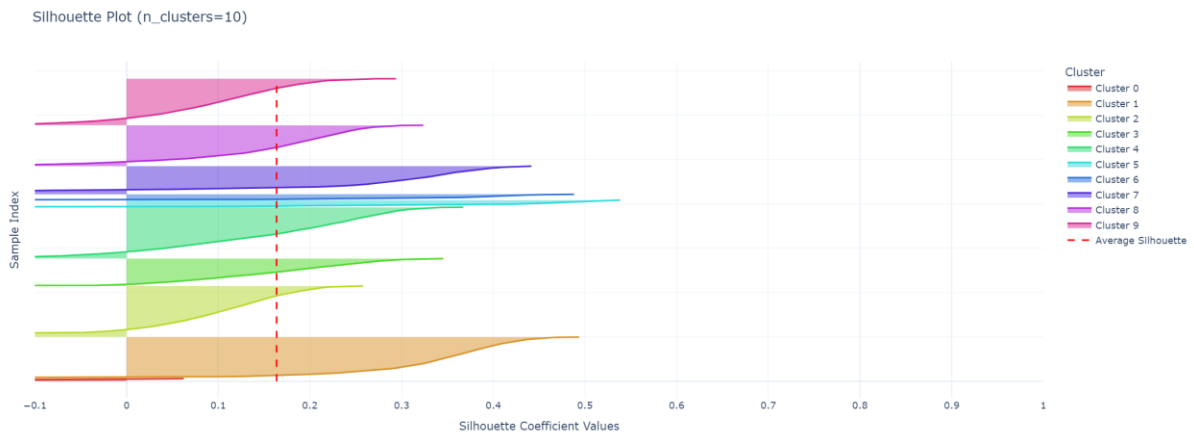


Figure 21 - K-Means + Hierarchical Silhouette

This method has shown the **best performance so far**.

3.1.8 SPECTRAL CLUSTERING

Based on a resource provided by Professor Ivo Bernardo (<https://scikit-learn.org/stable/modules/clustering.html>), we explored Spectral Clustering, **known for its ability to detect complex cluster shapes**.

This method constructs a similarity graph and performs dimensionality reduction before clustering, making it well-suited to data with intricate structures – even though it was not covered in our classes.

Final Outputs

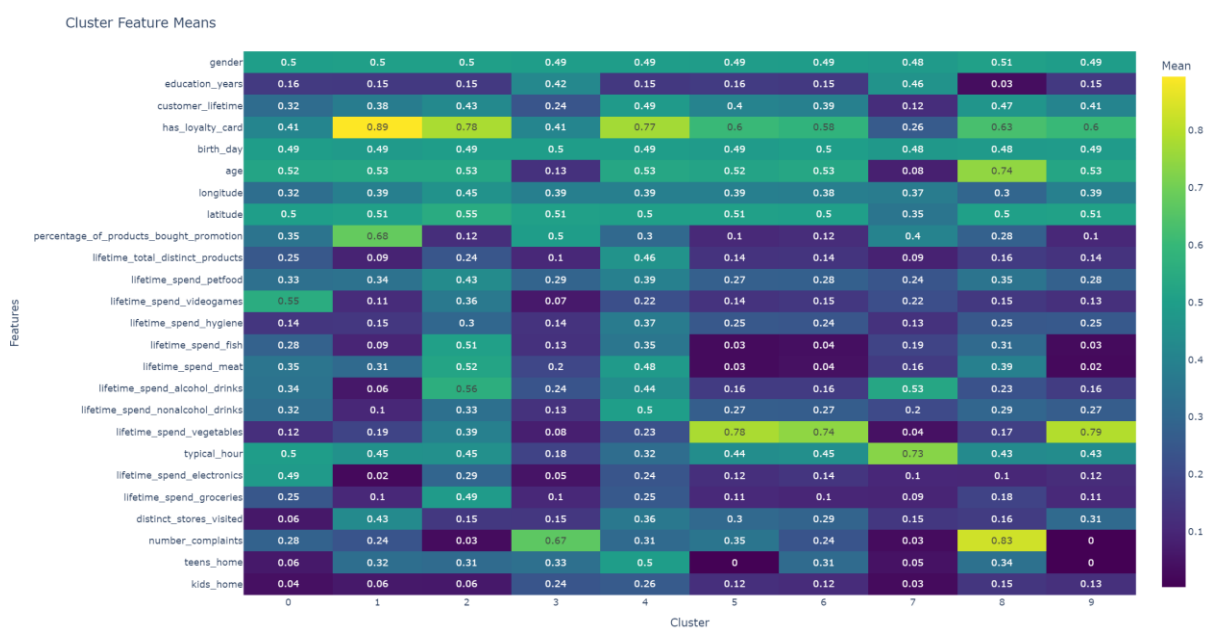


Figure 22 - Spectral's Scaled Mean Matrix

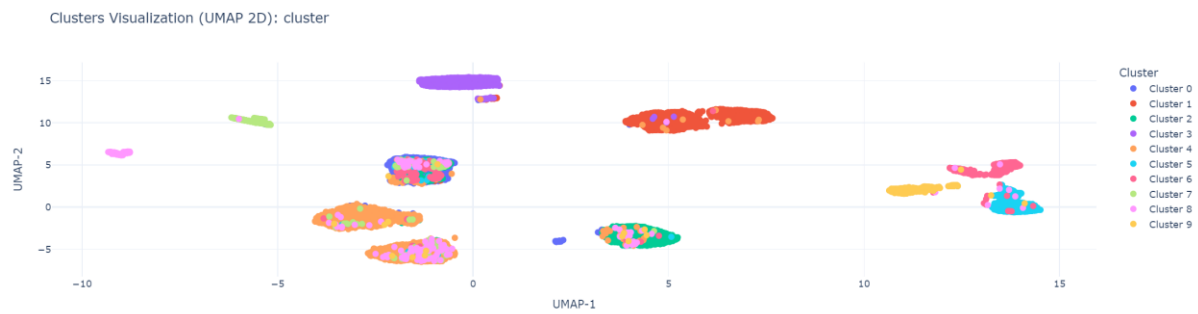


Figure 23 - Spectral's UMAP

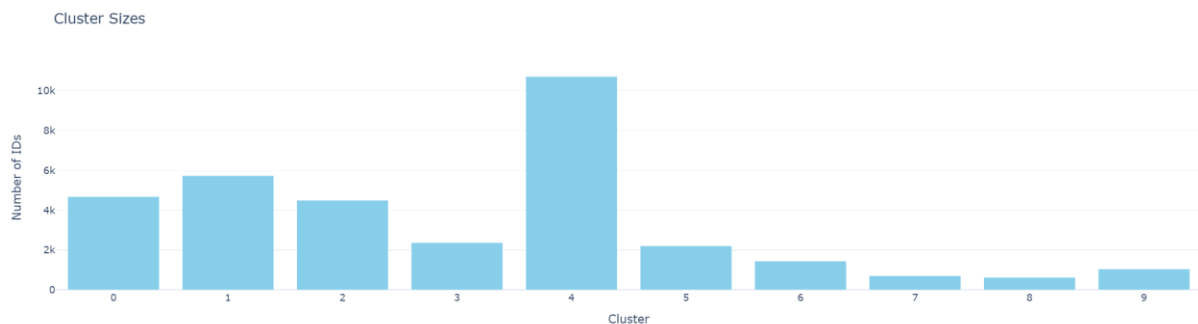


Figure 24 - Spectral's Cluster Sizes

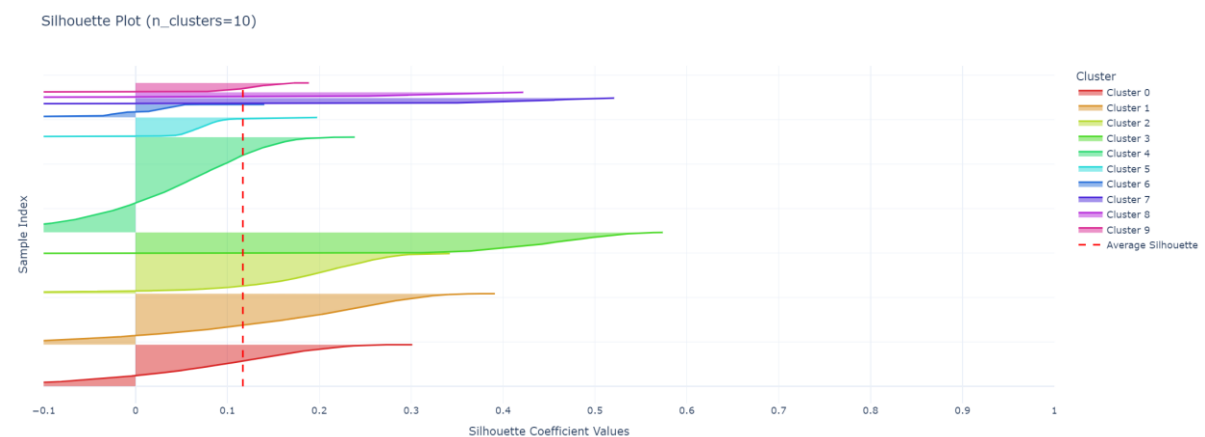


Figure 25 - Spectral's Silhouette

The Spectral algorithm yielded the best performance overall, according to the UMAP. However, it lacked interpretability, as it did not reveal strong distinctions between the resulting clusters.

3.1.9 CHAMPION CLUSTERING METHOD

Once multiple clustering approaches were evaluated, we determined that the **K-Means + Hierarchical Clustering** combination provided the most balanced and interpretable segmentation. This hybrid method leverages the efficiency of **K-Means for initial cluster formation** while integrating **hierarchical clustering to refine and group similar clusters**, leading to a well-structured and meaningful segmentation.

Why this clustering method?

- UMAP Visualization**
 This approach produced well-separated clusters, suggesting strong internal cohesion and meaningful distinctions between groups.
- Cluster Interpretability**
 The hierarchical step improved clarity, making it easier to allocate real-world meaning to each customer segment, as seen in the cluster matrix
- Algorithm Transparency**
 The sequential nature of the method allows for a deeper understanding of how clusters form and relate to one another.

Given these advantages, the clusters formed by the **K-Means + Hierarchical Clustering** approach will serve as the **basis for all subsequent analyses and interpretations of customer segmentation.**

Thus, the delivered ***final_csv_group9.csv*** contains all *customer_id* as well as their correspondent cluster computed by this method.

3.2 SUMMARY OF SEGMENTS AND THEIR CHARACTERISTICS

Cluster	Name	Key Characteristics
0	Makro Shoppers Store-based shoppers	- Cluster located near a Makro store - High spend across diverse categories (store designed to sell high quantities across all categories)
1	The Vegetarians Health-conscious or vegetarian individuals	- Very high spend on vegetables - Very low spend on meat, fish, and alcohol - Low complaints - Low purchase of promotional items (may prioritize health over price)
2	Large Households Family-oriented and variety-seeking shoppers	- Highest number of kids and teens at home - High spend on meat, alcohol, and non-alcoholic drinks (due to having a lot of people to feed) - High loyalty card usage - Slightly older age group (usually parents go shopping) - Buy the most distinct products (because of the age range at home, they have to meet everyone's needs)

3	Gamers and Tech Lovers Young, tech-savvy, entertainment-driven	• High spend on electronics and videogames • No dependents (kids or teens) • Low complaints • Moderate loyalty card use • Low spend on hygiene • Slightly lower education
4	Promotion Seekers Deal-hunting and price-sensitive shoppers	• High loyalty card usage • Highest percentage of promotional purchases • Visit many different stores (searching for the best deals) • Low number of distinct products bought (possibly cherry-picking deals)
5	Students Students or independent young adults	• Youngest cluster • Lowest loyalty card usage (maybe because they just don't care about that yet) • Higher education levels • Most purchases made later in the day (after classes) • Few or no dependents • High alcohol spending
6	Elderly Complainers Dissatisfied, disengaged	• Oldest group • Highest number of complaints (as a result of boredom) • Low to moderate spending across categories • Lowest education level (older times didn't require high levels of education) • Long customer lifetime • Moderate loyalty card usage
7	Money Grubbers Go around the system	• Complain a lot (to spend the least) • Buy mostly things in promotion
8	Normal Household Neutral Family	• Balanced spent in fish and meat • Older children • Use of loyalty card
9	Loyal Core Shoppers Usual Customers	• High loyalty card usage • Longest customer lifetime • Well distributed spent across almost all categories

3.3 SEGMENT COMPARISON

Feature	Lowest Value	Highest Value
<i>gender</i>	No significant changes	
<i>education_years</i>	6	5
<i>customer_lifetime</i>	5	9
<i>has_loyalty_card</i>	5	4
<i>birth_day</i>	No significant changes	
<i>age</i>	5	6
<i>longitude</i>	No significant changes	
<i>latitude</i>	No significant changes	
<i>percentage_of_products_bought_promotion</i>	1	4
<i>lifetime_total_distinct_products</i>	4 and 5	2
<i>lifetime_spend_petfood</i>	No significant changes	
<i>lifetime_spend_videogames</i>	7	3
<i>lifetime_spend_hygiene</i>	3	9
<i>lifetime_spend_fish</i>	1	8
<i>lifetime_spend_meat</i>	1	2
<i>lifetime_spend_alcohol_drinks</i>	4	8
<i>lifetime_spend_nonalcohol_drinks</i>	4	2
<i>lifetime_spend_vegetables</i>	5	1
<i>typical_hour</i>	7	5
<i>lifetime_spend_electronics</i>	4	3
<i>lifetime_spend_groceries</i>	4	8
<i>distinct_stores_visited</i>	3	4
<i>number_complaints</i>	5	6
<i>teens_home</i>	5	2
<i>kids_home</i>	5	2

This table serves only as a comparison in order to prove the previous cluster denominations.

3.4 VISUALIZATIONS FOR ANALYSIS

3.4.1 TRUE MEANS OF EACH CLUSTER

Cluster Summary (not scaled data):

cluster	0	1	2	3	4	5	6	7	8	9
kids_home	0.644211	1.029603	2.911303	0.286686	0.520304	0.129397	0.959624	1.617901	0.569031	0.956215
teens_home	0.410292	0.330304	1.895025	0.242954	0.951159	0.065327	0.937715	0.920706	0.930306	0.886084
number_complaints	0.884678	0.721698	0.863886	0.931908	0.663358	0.109548	2.790923	1.713688	0.123050	0.988045
distinct_stores_visited	2.316491	3.135542	3.047780	1.373372	3.955967	1.977746	2.139906	2.093854	2.078787	3.838949
lifetime_spend_groceries	23579.138844	10861.178242	20225.401246	23281.332685	10722.823219	12562.710668	19278.391236	21574.373218	46780.860426	31308.392157
lifetime_spend_electronics	7754.928187	2088.387767	5934.827271	9660.611597	658.745156	2071.623905	1774.636933	2309.520008	5086.210064	3275.708167
typical_hour	12.108772	13.859441	10.397046	15.545902	14.219613	19.754020	13.976213	10.109171	13.911406	13.406244
lifetime_spend_vegetables	1178.212330	1961.390007	775.859816	236.748753	471.551026	186.427220	411.428482	371.582413	979.353902	321.076223
lifetime_spend_nonalcohol_drinks	674.031471	405.626285	772.470911	499.031098	169.057792	288.479590	411.115806	248.045557	507.694076	690.116124
lifetime_spend_alcohol_drinks	1794.563052	442.672683	1430.938888	955.859475	187.263087	1261.795612	526.872300	673.488547	1448.775333	689.616905
lifetime_spend_meat	1322.772018	99.636328	1460.404558	897.156139	812.757319	423.061966	996.499218	643.489819	1360.840491	994.237007
lifetime_spend_fish	1463.629516	117.394360	1271.518547	873.577130	340.433822	585.082845	959.357746	571.002488	1612.982393	943.951799
lifetime_spend_hygiene	578.159274	704.656234	857.528992	325.265306	418.919437	378.819718	700.907668	450.035343	838.573215	1212.642109
lifetime_spend videogames	622.119167	213.992475	440.641980	1020.048267	207.822369	393.508723	224.388106	207.379730	595.974548	321.985151
lifetime_spend_petfood	265.399439	250.139128	350.780055	296.804017	305.739891	221.057790	306.932707	277.603017	390.820940	346.894917
lifetime_total_distinct_products	301.643198	87.151814	293.293258	130.572724	62.044584	61.023686	92.574335	83.643561	156.593958	256.106415
percentage_of_products_bought_promotion	0.187800	0.103713	0.212611	0.302552	0.683747	0.417401	0.297405	0.469051	0.119787	0.428264
latitude	38.740037	38.747568	38.747579	38.748021	38.747457	38.734748	38.748834	38.748241	38.751554	38.747330
longitude	-9.178853	-9.156721	-9.157563	-9.167915	-9.156536	-9.158967	-9.170074	-9.156028	-9.147709	-9.156302
age	52.186667	53.456539	54.400834	55.295692	54.758250	24.597990	69.367449	33.159597	53.231154	52.890691
birth_day	16.057544	15.686708	15.759166	15.817752	15.564165	15.634171	15.454147	15.918815	15.799435	15.736189
has_loyalty_card	0.382456	0.602944	0.686707	0.436022	0.873400	0.278894	0.618153	0.473369	0.764501	0.805445
customer_lifetime	10.125895	10.210369	11.530745	7.862909	9.604531	2.922161	11.925822	6.656615	10.763050	12.404378
education_years	6.698947	6.500704	6.328619	6.105021	6.257420	11.620603	4.205634	9.996533	6.521616	6.550390
gender	0.512281	0.488799	0.494526	0.504049	0.503113	0.486181	0.519562	0.497006	0.499240	0.492100

Figure 26 - K-Means + Hierarchical True Mean Matrix

This table is only a reference to check if the following visualizations act accordingly.

3.4.2 CLUSTERS' BOXPLOTS



Figure 27 - K-Means + Hierarchical Multi-Boxplots

The interactive visualisation, developed using **Plotly**, allows for the **exploration of the distribution of numerical variables between different clusters dynamically**. A dropdown menu was created to have an easy selection through the variables, which facilitates the comparison among the groups. Boxplots are displayed for each chosen variable, accompanied by a black **X** marker indicating the average for each group.

For instance, selecting the *lifetime_spend_vegetables* variable reveals that Cluster 1, representing the vegetarian group, has a notably higher average compared to the others. This approach provides a clear and efficient visual analysis of the differences between clusters.

3.4.3 LINE PLOT – CLUSTERS' PROFILES

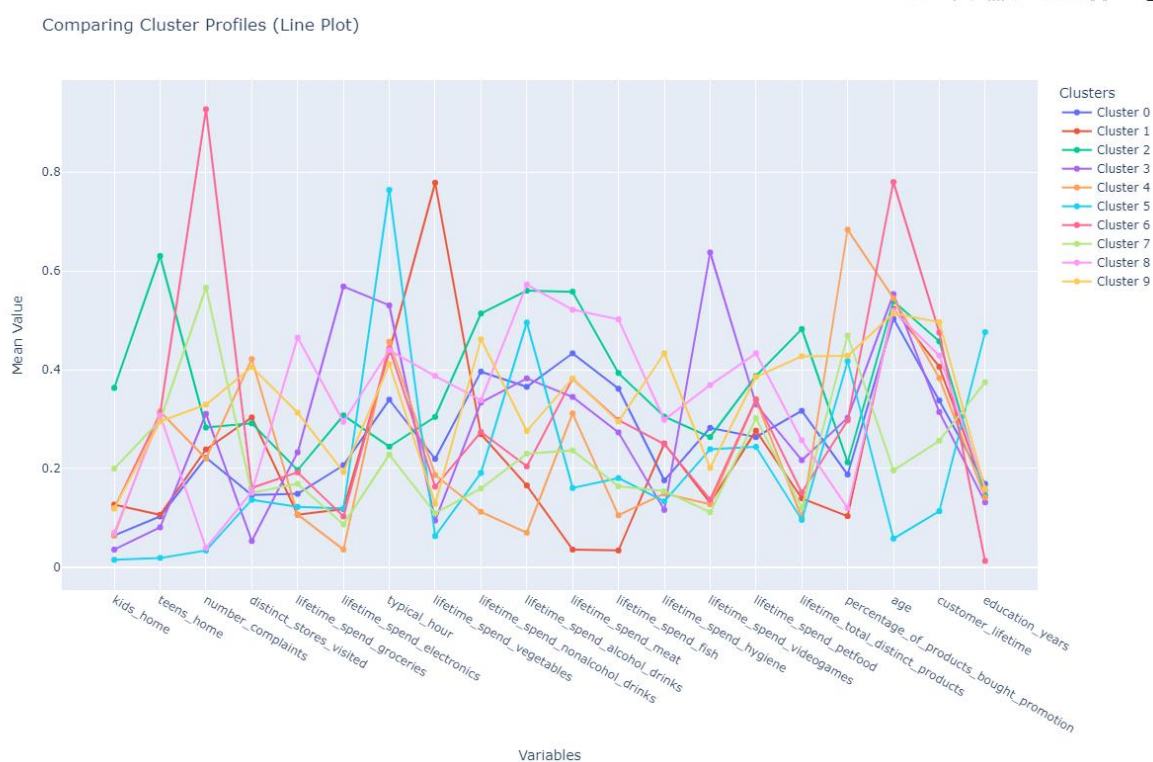


Figure 28 - K-Means + Hierarchical Line Profile Comparison

This line graph is useful for comparing the average profile of each cluster across different numerical variables. It shows **which clusters have the greatest variation between features** and identifies the **average value of each variable for each cluster**.

Plotly's interactivity enables hovering over the points to see the exact values, facilitating a detailed analysis. Additionally, the graph clearly shows which variables have the highest and lowest values by cluster, highlighting specific behaviours between groups. This approach is particularly effective for understanding how clusters differ in terms of key characteristics.

3.4.4 MOST AND LEAST VALUABLE CLUSTERS

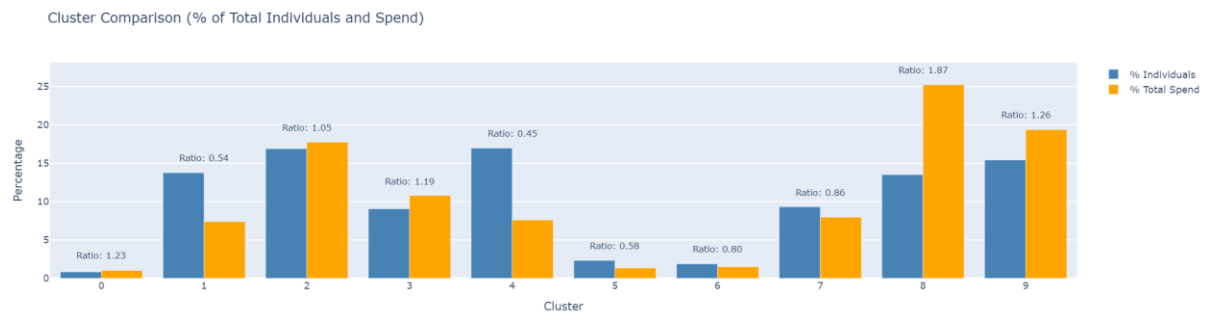


Figure 29 - K-Means + Hierarchical Ratio of Total Individuals and Spend

This grouped bar chart clearly visualises the representativeness and value of each cluster within the dataset.

The **blue bars** show the **percentage of individuals** in each cluster **relative to the total sample size**, enabling the understanding of the **size of each group**.

The **yellow bars** show the **percentage of total spending** attributed to each cluster when considering the **sum of all lifetime spend variables**.

By observing the relationship between these two metrics (cluster size versus spending contribution), **we can identify the most valuable clusters**, which are those whose spending contribution is proportionally greater than their size.

In this case, Cluster 8 (Normal Families) is the most valuable. On the other hand, Cluster 4 (Promotion Seekers) has the lowest ratio, with a spending contribution that is lower than its representativeness. This makes sense since this group tends to **look for the best deals**, thereby **reducing the company's profit margin**.

3.4.5 RADAR CHART

Radar Chart by Cluster

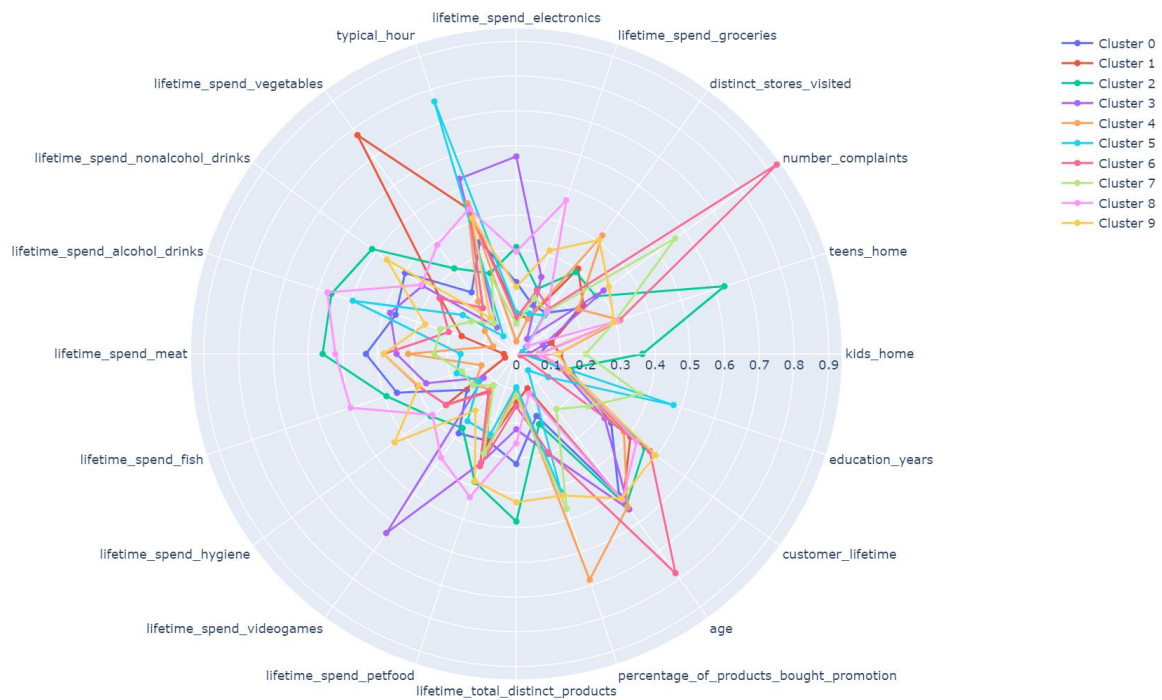


Figure 30 - K-Means + Hierarchical Radar Chart

Although this radar chart is arguably the least usefully informative of the visualisations presented, it provides an overview by highlighting which clusters stand out in each variable. This makes it possible to quickly and easily identify **which cluster 'leads' or 'stands out' in each characteristic** analysed, thereby reinforcing the logic behind the segmentation performed.

The circular representation facilitates general comparisons of average profiles, although detailed analysis is more challenging than with other types of charts.

3.4.6 DATABASE VS CLUTERS' AVERAGES

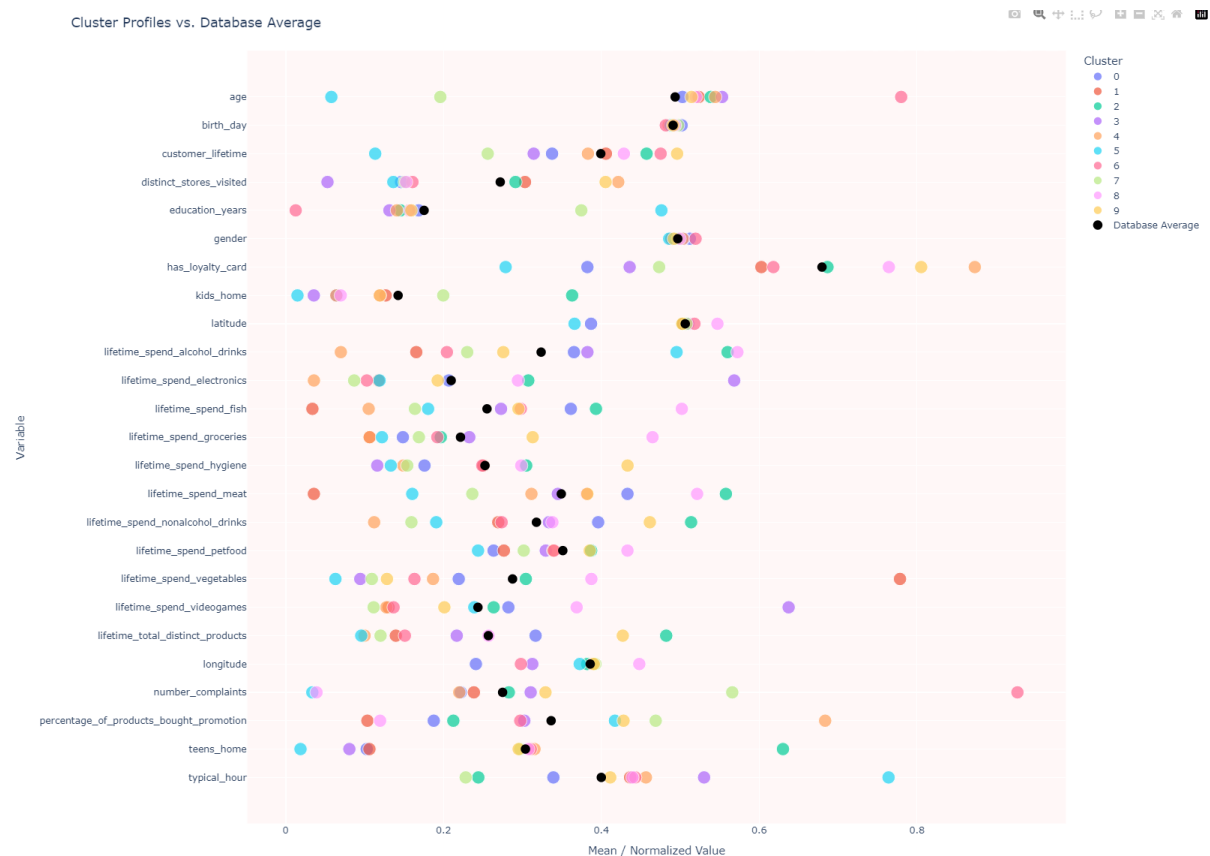


Figure 31 - K-Means + Hierarchical Database Vs Clusters' Averages

The purpose of this graph is to **compare the averages of the key variables** for each cluster with the **overall average of the dataset**. It helps understanding how each cluster is positioned in relation to the average profile of the base, highlighting specific variations for each variable.

We once more used *Plotly* for quick and succinct analysis of the clusters.

4 TARGETED PROMOTION

4.1 PROMOTION APPROACH BY SEGMENT (CLUSTER)

Given the segmentation and analysis that has been carried out, it is now **possible to identify the specific purchasing patterns of each cluster**. We then applied the **Apriori algorithm**, a well-known approach in recommendation systems, to each group in the segmented dataset. This enabled us to **determine the most suitable promotions for customers** in each cluster and **predict the products they are most likely to purchase**.

Using association rule mining and based on the customer characteristics in each cluster, we created **targeted promotions that were tailored to each customer profile**.

To extract relevant association rules, priority is given to those with the **highest lift**, followed by a **minimum confidence of 50%** and a **support threshold of at least 5%**, ensuring both the **strength and the popularity** of the rules.

4.2 MECHANICS USED IN PROMOTION

The **promotional mechanics** employed were primarily based on the **following strategies**:

1. Discounts
2. Coupons
3. Buy-one-get-one-free offers
4. Limited-time validity
5. Gift cards
6. Lifestyle discount programs

To design these promotions effectively, and as previously mentioned, we used **association rule application**. This enabled us to identify **meaningful purchase patterns within each customer segment**, leading to the results shown in the table below:

Cluster	Items	Support	Confidence	Lift
0	(Minecraft) (babies food)	This rule appears in 6% of all transactions in the cluster.	65.6% of those who bought Minecraft also bought babies' food.	The likelihood of buying babies' food increases by 4.64 if Minecraft is bought.
1	(shallot, asparagus) (carrots, tomatoes)	This rule appears in 5.2% of all transactions in the cluster.	57.2% of those who bought shallot and asparagus also bought carrots and tomatoes.	The likelihood of buying carrots and tomatoes increases by 1.29 if shallot and asparagus are bought.

2	(Minecraft, pet food) (Ratchet & Clank, babies' food)	This rule appears in 9.2% of all transactions in the cluster.	55.8% of those who bought Minecraft and pet food also bought Ratchet & Clank and babies' food.	The likelihood of buying Ratchet & Clank and babies' food increases by 1.2 if Minecraft and pet food are bought.
3	(cider, dessert wine) (white wine)	This rule appears in 6% of all transactions in the cluster.	81.6% of those who bought cider and dessert wine also bought white wine.	The likelihood of buying white wine increases by 3.2 if cider and dessert wine are bought.
4	(candy bars, cake) (tea, oil)	This rule appears in 7% of all transactions in the cluster.	53.8% of those who bought candy bars and cake also bought tea and oil.	The likelihood of buying tea and oil increases by 1.33 if candy bars and cake are bought.
5	(white wine, beer, cake) (salt)	This rule appears in 5.4% of all transactions in the cluster.	55.6% of those who bought white wine, beer and cake also bought salt.	The likelihood of buying salt increases by 1.65 if white wine, beer and cake are bought.
6	(candy bars, cake) (cooking oil, oil)	This rule appears in 7.8% of all transactions in the cluster.	56% of those who bought candy bars and cake also bought cooking oil and oil.	The likelihood of buying cooking oil and oil increases by 1.4 if candy bars and cake are bought.
7	(cake, asparagus) (cooking oil, oil)	This rule appears in 6.56% of all transactions in the cluster.	57.86% of those who bought cake and asparagus also bought cooking oil and oil.	The likelihood of buying cooking oil and oil increases by 1.5 if cake and asparagus are bought.
8	(spaghetti, frozen smoothie) (champagne, fresh tuna)	This rule appears in 6% of all transactions in the cluster.	58% of those who bought spaghetti and frozen smoothie also bought champagne and fresh tuna.	The likelihood of buying champagne and fresh tuna increases by 1.4 if spaghetti and frozen smoothie are bought.
9	(ham, cologne) (barbecue sauce, oil)	This rule appears in 6% of all transactions in the cluster.	57.9% of those who bought ham and cologne also bought barbecue sauce and oil.	The likelihood of buying barbecue sauce and oil increases by 1.3 if ham and cologne are bought.

The following promotional strategies were developed by **combining insights from association rules with cluster interpretations** derived from the graphs and visualisations in the analysis.

Cluster 0	Make a purchase over 100€ and save 25€ on your next purchase!
Cluster 1	Live healthy, save more! Spend over 100€ on healthy products, vegetables, fruits, juices and many more and receive a 15% discount in the electronic section! This offer is limited!
Cluster 2	Per 20€ spent, receive 1 coupon. Once you collect 15 coupons, enjoy a 50% discount on all meat products!
Cluster 3	Spend over 100€ on groceries and vegetables and get 20% off on any electronic product. But wait, that's not all! Get 1 video game for FREE!
Cluster 4	Weekly Exclusive Offer! Buy 2 juices and get 1 dessert for FREE! Only active this week!
Cluster 5	Refer a friend and enjoy a 15% discount on your next purchase. Your friend also gets 5% off their first purchase!
Cluster 6	Rewards make us happy! Spend 20€ and receive 2 exclusive coupons! Later, redeem them for 15% discount on sweets and baked goods! Don't delay, act fast!
Cluster 7	Exclusive promotion just for you! Buy a cake and get 50% off on party stuff (cups, candles, napkins...).
Cluster 8	Don't miss your one and only chance to grab this deal: Buy spaghetti and tuna and get 30% off on champagne!
Cluster 9	Exclusive promotion just for our loyal customers! Spend over 30€ and receive 3 coupons. With only 10 coupons collect free pair of headphones!

5 CONCLUSION AND RECOMMENDATIONS

The **performance of clustering algorithms** is heavily influenced by the **quality of the pre-processed dataset**. Since most clustering techniques rely on distance-based measures, it is crucial to handle outliers carefully, treating them apart during the preprocessing and clustering processes to prevent distortion, but reintegrating them into the post-clustering analysis to ensure a comprehensive understanding of all customer types. Similarly, binary variables can disproportionately affect clustering outcomes due to their extremity. As such, these variables are better excluded during clustering but reintroduced during analysis to enrich the interpretation of the resulting segments.

To **assess the quality and structure of the clustering solutions**, we applied **dimensionality reduction techniques such as UMAP for visualization**, alongside **quantitative metrics** like the **silhouette score**. Interpretability was a key objective, and visual tools proved valuable in achieving this. By visualizing the clusters in two dimensions, we were able to uncover previously hidden patterns – most notably, the emergence of a new cluster representing high-value customers (Makro Cluster).

Despite testing multiple clustering algorithms, no single method consistently yielded optimal segmentation across all criteria, however, we **ended up opting for K-means together with Hierarchical Clustering**. To deepen the analysis, we compared the mean of each variable within a cluster to the overall dataset mean. This allowed us to define clear, distinct customer profiles based on behavioural and demographic patterns.

Following the identification of these clusters, we applied an **association rule process** to extract actionable insights and generate tailored marketing strategies. These rules help design **personalized promotions that resonate with the preferences of each segment**, ultimately aiming to **improve customer satisfaction and attract new clients**.

We **strongly recommend this methodology** because it delivers both analytical depth and practical value. The integration of **clustering and association rule application provides a robust framework for understanding customer behaviour and supporting data-driven decision-making**.

Furthermore, the **interpretability achieved** through visualization enhances **communication with stakeholders** and **ensures that insights can be effectively translated into strategic actions**.